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Cloudbusting Experiment

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BASIC INFORMATION:

OROP ARIZONA 1989:

A Cloudbusting Experiment To

Bring Rains in the Desert Southwest *

by James DeMeo, Ph.D.**

Cloudbuster *Icarus,* a large remotely-controlled apparatus constructed by James DeMeo in 1978, shown h on the banks of the Colorado River during the *OROP Arizona 1989* project.

* Published in *Pulse of the Planet #3,* Summer 1991, p.82-92. A previous version of this article was serializ *Journal of Orgonomy*, 23(2): 271-272, November 1989, 24(1):111-124, May 1990, and 24(2):252-258, No 1990.

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Acknowledgements: A number of individuals physically assisted with these desert operations, taking all tl associated with cloudbusting in a hostile desert environment. Special thanks to Theirrie Cook, who particip almost all of these operations, and to Donald Bill, Argyro Collins, Dennis Collins, Alberto Foglia, Sybilla I Stephan Muschenich and Vittorio Nicola for their help and assistance. Thanks also to Richard Blasband, R Nunley, and Shafia Lave, who assisted with additional cloudbusters in the San Francisco Bay area and in K several occasions. Funding for these field expeditions was provided by the American College of Orgonomy small grant generously provided by Mr. Lou Hochberg. The first of the experiments described in this paper 1989, were video-recorded by a German video documentary team consisting of Jurgen Fischer, Beate Freil John Trettin -- click here for a special note on the video documentary. *OROP Arizona* was conceived by Dr the mid-1980s, as a systematic evaluation of the cloudbuster's *desert-greening* capabilities, and as a test of prior observations and claims as previously published in the 1950s. Preliminary field work was undertaken Blythe, Arizona region in the Fall of 1988, showing good results, and the full project was finally carried fo

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1989, as organized by Dr. DeMeo through the *Orgone Biophysical Research Laboratory*.

I. Preliminary Report on Five Cloudbusting Expeditions into the Desert Southwest, May-September, 1989

Background

Several years ago, the increasing pace of desert expansion, drought, and general atmospheric chaos prompted me the development of a Desert Research Center in the arid portion of the American Southwest, with the goal of der a long-term desert greening effect using the Reich cloudbuster. During the years 1952-1956, Wilhelm Reich inve basic energetic-atmospheric functions related to drought and desert formation, and demonstrated clear desert gre effects during his cloudbuster operations near Tucson, Arizona (1, 2). His early experiments provided the basic g principles for this more recent desert research (3,4). As a preliminary step in the development of a long-term des project in the American Southwest, it was important to determine how much influence separate short-term cloud operations could have on the desert atmosphere in an exceedingly dry area. In the summer of 1989, this experim program of desert cloudbusting was conducted. Five seperate field experiments were planned and announced in undertaken by teams of workers using portable equipment, and recorded in extensive field observations. Nationa Service (NWS) data were used later to analyze the geographical distribution and intensity of atmospheric effects

The operations proceeded as follows: A transportable cloudbuster was driven to a very dry part of the American and set up for periods lasting a few days to a maximum of one week. After each operation, the research team wit cloudbuster left the area, completely suspending operations until the next expedition aproximately 3-1/2 weeks l allowed natural weather conditions to reestablish themselves. One experiment with this quick, mobile cloudbusti previously undertaken in the desert Southwest in 1988, with much success (3). A site along the Colorado River, miles north of Mexico on the California/Arizona border was chosen for the experiments because of its easy acce abundant water and central location within one of the driest parts of North America. A second cloudbuster locate Francisco Bay area, operated by Richard Blasband, was used on two separate occasions to assist these desert ope with a third, very simple cloudbuster located in Kansas for similar purposes. The Kansas portions of the operatic undertaken to enhance the downstream rainfall effects of any energetic potentials propagating toward the central the desert Southwest.

The five operations took place at approximately monthly intervals, beginning the first week of May, 1989. A lett the dates and purposes of the desert cloudbusting expedition was sent to the National Oceanographic and Atmos Administration offices in Washington, DC, and telegrams for documentation were dispatched to those offices as prior to the onset of each individual operation.

Operation #1: May 8-10, 1989

This first operation began at a time when daytime temperatures were peaking at around 100-105°F. There was a possibility of slightly lower temperatures, but certainly no rains or significant clouds were expected. (May and Ju driest months in the desert Colorado River basin region.) Plate 1 shows the thick atmospheric haze -- what Reich DOR (for deadly atmospheric orgone) -- at the start of this particular desert cloudbusting operation. Reich was th natural scientist to observe that such atmospheric haze possesses an energetic component which acts to block the development of clouds and rain.

Plate 1: A harsh desert landscape prior to cloudbusting operations, near Blythe, California. Note the heavy in the atmosphere, which obscures the nearby mountains. Shade temperatures at the time of this photo (11: were around 99°F, with a 15% relative humidity, typical oppressive summertime conditions for the area.

After one day of operations, thunderstorms blossomed to the south of the cloudbuster, over northwest Mexico. A

days of hard work under the hot sun, streams of energy and moisture began to enter the dryland Colorado River l from both the west and the south. The westerly energy stream was composed of what classical meteorology calls jet"; after observing this shifting weather pattern, one puzzled TV meteorologist said, "It just can't do that" -- the "frontal boundary" or air mass with a sufficient temperature differential to "push" the jet stream southward into And yet, the energy stream danced southward many thousands of miles from its original location near Alaska. Tl southerly energy stream was composed of the "sub-tropical jet," and it moved north into our area from the Gulf California.

These streams of energy and moisture appeared as "fronts" on the weather map and were seen clearly as a long l wispy, high-altitude cirrus clouds standing out against the desert sky (Plates 2 and 3). When the energy streams hazy, stagnant DOR conditions in the lower atmosphere diminished, even though the clouds demarcating the ene were many thousands of feet in elevation. When the energy streams moved away, stagnanant DOR conditions w clear demonstration these wispy clouds represented much more than just high-altitude "jet streams" of mechanic air.

Plate 2: Cloudbuster Icarus at work along the banks of the Colorado River, under broiling, DOR haze conc

Plate 3: Passage of the Galactic orgone energy stream (southerly "jet") overhead. Note the clearing of low- haze conditions with the passage of this energy stream, one of several indications that it is composed of mc mechanically-driven, high altitude winds.

As these two energy streams moved inland, one from the northwest, one from the south, they gradually approach superimposed far to the north in Montana, causing gusty winds and heavy rainstorms. Even though this superim sequestration was happening far away, the spiraling motion "sucked" much of the DOR-laden weather out of the basin, toward the center part of the storm system. With this effect, the cloudbusting operation was terminated lat evening of May 10. The next morning saw an absolutely crisp and sparkling sky, completely free of the previous DOR haze, as seen in Plate 4. Temperatures that day only reached about 70°F! Crisp, cool, sparkling conditions over the Arizona/New Mexico region for weeks, bringing widespread rains in the surrounding mountain areas. T spread northeast into the Great Plains and Midwest.

Plate 4: A moving, pulsing atmosphere in the desert, also close to Blythe, California, after completion of tl cloudbusting operations in early May, one of the driest times of the year. Note the greater transparency of t atmosphere, and the developing cloud cover. Shade temperatures at the time of this photo (11:00 AM) wer 67°F, with a 50% relative humidity, very unusual crisp and clear conditions. Contrast this photo with that g Plate 1, which shows the same landscape prior to cloudbusting operations.

The precipitation resulting from this first operation is documented in Figure 1, which shows the NWS daily prec averages for April and May of 1989 from 424 different weather stations in the southern half of both California a and from the entire state of Arizona (5). As shown in Figure 1, increased rainfall was observed across this very l mostly arid geographical basin and range territory immediately following the cloudbusting operation of May 8-1 previous pulse of moisture on April 21-27 appears in Figure 1, the result of a coastal storm that moved into Calif lacked enough energy to move inland over the high mountains (4000'-8000' elevation). The cloudbusting operati 8-10, however, was followed not only by increases in moisture and rains in coastal and southern California but a moisture that moved far inland and upslope, into the basin and plateau regions of Nevada and Arizona where sig rainfall developed. The lag time between the onset of cloudbusting operations and the time of first rainfall was s (around one day) for the wetter coastal regions of California, and longest (around four days) in the arid desert ba These data do not reflect thunderstorms blossoming in Mexico only one day after operations.

This kind of atmospheric response to cloudbusting during May, one of the driest months in the Colorado River E was remarkable, most gratifying, and in complete agreement with the results of previous desert cloudbusting ope undertaken in August of 1988 (3).

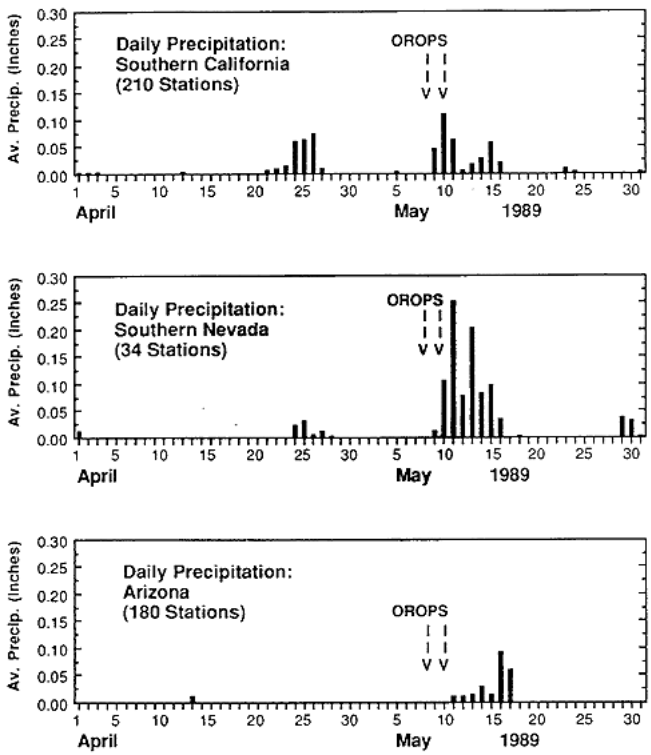

Figure 1: Daily Precipitation, average daily values for 424 weather stations in southern California, souther and Arizona (6).

Operation #2: June 5-8, 1989

The second desert cloudbusting operation proceeded at a time of increased aridity and temperature. Widespread burning hampered visibility at the cloudbusting site and appeared to "choke up" and reduce the responsiveness o atmosphere to the cloudbuster. In addition, a French nuclear bomb was tested in the western Pacific. The atmosp sluggish, immobilized, and had a "resigned" quality. After two frustrating days of work with the cloudbuster, the observable change was a slight reduction in peak air temperatures. An assisting coordinated draw from a second located in the San Francisco Bay area was then begun, and this provided the additional energetic push needed to moving. Within a short time, a low-pressure storm system swept southeast from the San Francisco Bay area tow location on the Colorado River, on a line directly between the two cloudbusters. This storm system dropped abu precipitation on the southern Sierra Nevada Mountains, an unlikely occurrence for the season. Plate 5 shows the outbreak of these mountain thunderstorms. Operations were terminated shortly after the onset of these rains, but showers continued to blossom in the same remote, dryland area of southern California over the following days, p farther inland, into southern Nevada and Arizona.

Plate 5: Thunderstorms develop over the southern Sierra Nevada Mountains following the cloudbusting op early June.

Operation #3: July 3-5, 1989

This operation took place at a critical time. Two nuclear bombs were detonated underground in Nevada the prev Also, two of the three Palo Verde nuclear power plants, near Phoenix, were being loaded with nuclear fuel, whil had a recent emergency shut-down. Nuclear refueling and emergency shut-down procedures are well-known as ' procedures, releasing relatively large quantities of radioactivity into the local atmosphere. These events produce *oranur* reaction in the Southwestern U.S., an overexpanded, overexcited, highly charged atmospheric energetic c with virtually no cloud cover or rains (6). An unusual clockwise-rotating low pressure system dominated the are addition, this early July cloudbusting operation was scheduled to start just before the natural annual transition fr to the wettest period of the desert Southwest, the so-called Arizona "monsoon" season which is characterized by thunderstorms. We predicted our operations would result in an earlier onset of the monsoon with a greater than r quantity of rain.

Operations began on a day with shade temperatures over 110°F, and relative humidities at or less than 15%. Aft operations, some very large thunderstorms appeared to the south of the cloudbuster over northwest Mexico. A ve high-altitude streaming movement from south to north also developed but failed to trigger any storms, until it rea western Nebraska where thunderstorms blossomed. Conditions worsened as the draw continued. By the second c operations, we recorded a shocking 123°F shade temperature, the hottest in the entire U.S. for that day. Oranur c intensified, making work near the cloudbuster almost unendurable. Satellite images continued to show no storms the vicinity, except in Mexico and southeast Arizona.

Cloudbusting operations were terminated after the third day of widespread high temperature, high-pressure cond which threatened to spread northeast, and trigger drought across the Great Plains and Midwest. However, a final cloudbusting draw was made from both cloudbusters, on the Colorado River and in the San Francisco Bay area, good westerly flow to push this dome of hot air eastward. This proved successful, and over the following week t hot air moved eastward across the entire U.S. and the Atlantic Ocean, eventually appearing in Western Europe, v "stuck" for a period before finally dissipating. As the hot, agitated air and energy moved eastward out of the des Southwest region, clouds and rains filled in behind it, expanding northward from Mexico and southern Arizona, increasing in tempo over subsequent weeks. The monsoon season had arrived.

Operation #4: July 31-August 3, 1989

This operation began at a time when the southerly flow of moisture into the desert Southwest had already been u several weeks, with widespread thunderstorms and rainshowers occurring every other day. The goal this time wa

increase rainfall to above normal amounts. DOR levels were still markedly elevated in some regions despite the several forecasters indicated the really heavy "monsoon" rains had yet to appear. An easterly flow of air from ou Pacific Ocean was also reported to be hindering rains in the area, putting a "cap on instability"; this was contrary expectation, given the air mass' origination over the moist ocean. More was learned about the stagnant DOR natu of these Pacific air currents and will be mentioned shortly.

After one day of operations, a large complex of thunderstorms developed just to the east of our site. The Nationa Service reported significant showers in a few surrounding areas, but no widespread rains were observed. In the e Pacific Ocean, Hurricane Gill began churning northward along the coast of Baja, and the counter-clockwise spir around this storm spread moisture inland up the Colorado River Valley. By the second day of operations, relativ were at 69%, although widespread rains failed to develop from this moisture. By the third day, Hurricane Gill co fizzled out, and all clouds evaporated across the region.

We were very puzzled and frustrated by this turn of events, and acutely felt the need to better understand the natu Pacific air currents, which could sap the strength from approaching hurricanes and squelch rains far inland, even humidities and temperatures prevailed. This cloudbusting operation was terminated, and no assisting operations this time from the San Francisco Bay area. Moisture continued to stream into the desert Great Basin, however, a days later, an incredible series of events occurred, evidencing a widespread break-up of the desert armor.

1. A "mild" earthquake, magnitude 5.2 on the Richter scale, and some highly unusual "earthquake weather" occu of the San Francisco Bay area. Attention was drawn to the connection between the earthquake and the strange w primarily because a local TV meteorologist spent considerable time denying the existence of such a connection. satellite images, a unusual pattern of thunderstorms could be
seen radiating outward from the earthquake zone, like spokes on a wheel, to the north, east, and south. Heavy rai strong winds were produced from these storms in southern Oregon, Nevada, California, and Arizona. Rains, in f persisted for days in areas where, as one TV weatherman remarked, "It ought not to be raining."

2. Yuma, Arizona, received two years' rainfall in one week, and other Arizona and southern California stations li experienced heavy rains, and some flooding of low-lying desert topography was reported. More rains broke out entire Great Basin region in subsequent days and continued through the month. By the end of August, the region cloudbusting site had received a whopping 500% of its normal summertime rainfall, as seen in Figure 2.

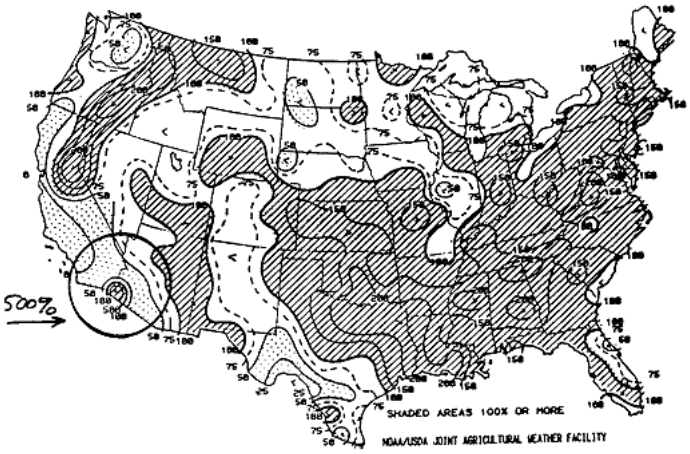

Figure 2: Map of Percentage of Normal Precipitation for the U.S., June through August, 1989. Shaded are 100% to 250% of normal rains. Note the region of 500% of normal rainfall in the desert Southwest. The cl was located at the center of the large circle, close to that region of greatly increased rainfall.

Operation #5: August 28-September 1, 1989

This operation began under hazy, DOR-infested, cloud-free conditions. After one day of operations with the clou partly cloudy conditions developed across the Colorado River Basin area, with scattered showers across Arizona in the mountains. A large complex of thunderstorms developed in Mexico, south of Yuma, Arizona, and also ne cloudbuster. Given this rather immediate response to the operations, it was decided to initiate a second draw fro coastal region just north of Los Angeles, hopefully to bring Pacific moisture more directly to the inland desert re cloudbuster was moved to Moro Bay, on the coast, and grounded in the Pacific Ocean, drawing westward direct large fog bank. After one-half day of work at this site, the operation was terminated, and we returned home to ob subsequent events.

The results were delayed, but significant. A tropical depression in the southern Pacific, about one thousand mile slowly moved northward off the coast of Mexico. Over a period of several days, this depression developed into Storm Octave and later became Hurricane Octave for a short period. This storm continued northward off the coa but soon weakened and spread rains over a large area. There was enough momentum, however, to bring these ra southern California. This entire slow process took about 10 days. On September 15-17, highly unusual, unseaso

moved inland across southern California. This was the first significant precipitation in two years for some parts c California. The atmosphere was refreshed and cleaned of its DOR-infested, hazy characteristics. The moisture w welcomed by firefighters and foresters, who were previously sounding an alarm about the dry conditions. A secc pulse of moisture occurred in the same area in late September.

Summary of Part I

With the exception of the first cloudbusting operation, NWS weather data have not been reviewed to provide "of documentation for the field observations described here. Based on past experience, there is good reason to believ be fully corroborated and strengthened by such a review and data analysis (NOTE: These data are now presentec report which follows on page 21). Certainly, these desert cloudbusting experiments have provided a wealth of ob plus many new and important points worthy of future study and research:

1. Preliminary indications suggest, as indicated in Figure 2, some areas of the desert southwest (southern Califor southern Nevada, and Arizona) were unusually wet, receiving up to 500% of normal rains; other parts of the sam remained quite dry, however, receiving less than half their normal precipitation. This Figure suggests that DOR- air was sequestered in some areas but not in others. (Note: Figure 2 does not include data on the rains following cloudbusting operation, which was also followed by at least two major pulses of rainfall in southern California.)

2. Fire danger in the Western states dramatically lessened in 1989, particularly after rains in August and Septeml major breakthrough occurred. For example, by September 15, 1988 (a record dry year in the U.S.), forest fires cl nearly 4.1 million acres in the Western states, with a total of over 5 million acres burned for the year. As of Sept 1989, following the series of desert cloudbusting operations, only 1.4 million acres were lost, with a very low fir most parts of the U.S. (7).

3. Stagnant, DOR-infested air was repeatedly observed moving into coastal southern California and the Southwe Great Basin from off the Pacific Ocean itself. On several occasions, measurements of cool, low-humidity winds from the open ocean were made right on the Pacific shore. While normally attributed to "cool ocean currents," th ocean winds appear to involve other energetic factors. They were contaminated, for example, with a thick DOR at times completely obscured the horizon, similar to the DOR haze in the heart of the desert. Plate 6 shows DOR inland from the open ocean as seen from an altitude of several hundred feet. Usually identified as a "cooled mari "dry fog" with a characteristic temperature inversion, the DOR "inverted" air was literally followed as it blew in mountain passes, into the dry inland valleys of California and Arizona. When such hazy DOR was first observed desert field station along the Colorado River, it was thought to be Los Angeles DOR air pollution; another time, smoke from agricultural fires; then, DOR blown out to sea from the Mexican deserts, returning inland to the sou California coast. In the final analysis, however, this DOR probably originates far out over the Pacific Ocean, pos the deserts of Asia, as discussed in a prior paper on "Desert Expansion and Drought" (4).

Plate 6: The Pacific DOR haze layer, mechanistically termed the "marine inversion" or "dry fog" layer. Th humidity air is moving onshore from a dry zone over the eastern Pacific Ocean, into a largely coastal dryla eventually merges with and feeds the DOR haze layer of the inland deserts.

4. While the exact nature of the 1987-1988 drought crisis has yet to be understood, Figure 3 suggests there are at cyclical components. The Figure first identifies a regular summertime increase in the area of severe or extreme d within the U.S. This cyclical aspect is a result of the "normal," "dry season" climate affecting the California coas and appears to be related to the dry Pacific Ocean DOR layer, discussed above. However, in the summer of 1987 "seasonal" component expanded from a yearly maximum of 10% to over 35% of the entire U.S. being affected b extreme drought conditions. The reasons for the enormous expansion of drought during the late summer of 1987 mystery. Part of the answer, however, may lie in the movement of a greater than "normal" quantity of DOR from Ocean source into the U.S. This span of dry, DOR air, moving east-northeast from the Pacific Ocean inland acro Rocky Mountains into the Central U.S., can often be observed as a huge swath of cloud-free atmosphere on satel one striking example of which was previously published in the *Journal of Orgonomy* (3:107).

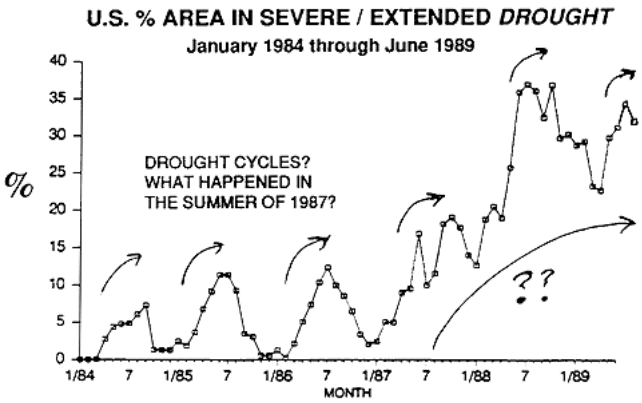

Figure 3: Percent of U.S. with "Severe or Extreme Drought" conditions. National Climatic Data Center, N

5. The desert Southwest exerts a controlling influence over rainfall in those wetter regions immediately adjacent DOR builds up and intensifies within the desert basin regions and "spills over" into adjacent wetter regions, drou waves, and forest fires are triggered. On the other hand, as rains break out within the desert region itself, the DO mass is reduced, and rains subsequently appear and are more easily maintained in those wetter borderlands. In th the summer 1989 cloudbusting operations in the desert Southwest appear to have been helpful, perhaps crucial, i eliminating drought conditions in parts of the Great Plains and Midwest "downstream" from the Southwestern d seen in Figure 2 above, for the summer months of June, July, and August of 1989, most parts of the U.S. east of Mountains experienced rainfall totals of over 100%, with many areas exceeding 200% of normal precipitation. T Plains and Midwest have clearly emerged from their prior drought condition, and rainfall totals in most of those states have increased above the deficit levels of prior drought years. Figure 4 emphasizes the unusual nature of tl 1989 rainfalls, which increased in magnitude and scope to levels not generally occurring in the U.S. since Januar Indeed, the problem in some areas of the U.S. during the summer and fall of 1989 was too much rain. Although optimal situation, this was clearly better than the catastrophic drought and fire conditions prevailing in 1988. Re obtained NWS rainfall data indicate the coastal and interior parts of central California received over 500% of its September rainfall (*Weekly Weather and Crop Bulletin*, October 11, 1989, pg. 12).

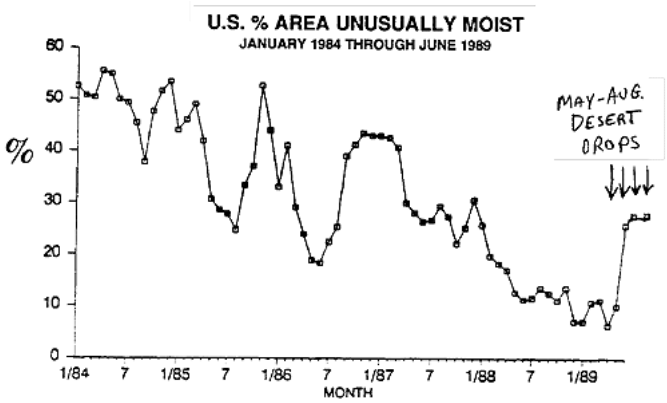

Figure 4: Percent of U.S. with "Unusually Moist" conditions. Note the dramatic increase in moist regions May-August cloudbusting operations.

6. Observations made during these experiments support the hypothesis relating earthquakes and weather. In parti earthquake of only moderate magnitude was followed by the geographically related genesis and movement of st a previously "stuck," immobilized atmosphere.

7. The relationship between underground nuclear bomb testing and atmospheric phenomena needs further resear study. Soaring summertime temperatures were observed in the open desert shortly after a series of such tests. Ad systematic observations suggest a tendency for rainfall-diverting high pressures to build in the Pacific Ocean off California coast following underground testing in Nevada. Similarly, an increasing buildup of DOR-laden air, fo drought or severe storms, has been observed in the Great Plains also following testing in Nevada. These observa keeping with the recently published findings of Kato (8) and Whiteford (9), which can only be explained on the nonmechanical, energetic process.

All the above observations and findings reinforce strongly the author's feeling that a more coherent and focused

atmosphere from an orgone-energetic viewpoint is vitally necessary. The connections between desert and drougl global nature of various rain-inhibiting "inversion" phenomena need rigorous study (4). Careful documentation c atmospheric effects of underground nuclear testing and practical solutions to the problem of desert expansion are essentials for our collective survival. This cannot be left to mechanistic meteorologists, who generally deny any relationships, or that any problem exists.

II. Data Analysis of Five 1989 Desert

Cloudbusting Experiments

National Weather Service (NWS) precipitation data were obtained from a total of 424 different precipitation mea gauges located in Southern California, Southern Nevada, and Arizona (10). Figure 5 shows the areas from which precipitation data were gathered, and the primary location of the cloudbuster during the experiments. The NWS presented here indicate a significant increase in precipitation within the large dryland basin region following clou operations. It is recalled from the preceding report that one region, immediately adjacent to the site where the op cloudbuster was located and about 100 miles in diameter, received up to 500% of the normal summer rainfall (se The data presented here indicate that precipitation amounts following the cloudbusting operations generally incr much larger territory within the arid desert basin. This increased rainfall was not uniformly distributed, however concentrated according to the wind and cloud characteristics which developed after the operations were under w

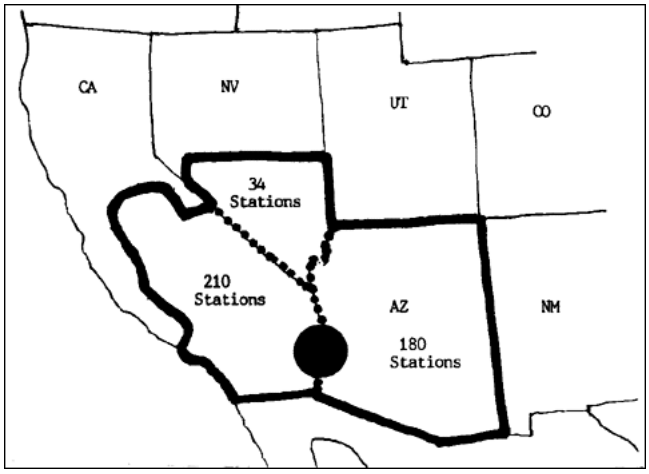

Figure 5: The area of the Southwestern USA used for the precipitation analysis is outlined, with the numb gauges given. The cloudbuster was operated from a site on the Colorado River, marked with a large dot.

The first of the five 1989 cloudbusting operations, for example, took place in early May and was characterized b west-to-east energy streaming ("jet stream") into the area and the inland movement of moist Pacific winds with f activity. These westerly winds brought moisture and rainfall first to the mountain areas of Southern California, i Los Angeles basin region, and second to the arid Colorado River Valley and highlands of Arizona and Southern Figure 6, which gives average daily rainfall data from 210 different NWS gauges in Southern California, shows relationship. The first cloudbusting operation of early May, characterized by westerly winds, was accompanied a by a clear peak in southern California rains; the other four operations, which did not have as strong westerly win not. Instead, the predominant energy streams and moist winds during and immediately following the last four 19 cloudbusting operations of early June, July, August, and September were characterized by a generally south-to-n movement from the equatorial Pacific Ocean, northeast across Baja and the Gulf of California, northward throug Colorado River Valley, and then north and east into Arizona and the Nevada highlands. In these latter four cases frontal activity was minimal, and isolated inland thunderstorms predominated; moisture moving north through th California, therefore, did not significantly affect Southern California, but rains did develop in Arizona and South Figure 7 documents this observation and summarizes NWS data from 214 different rain gauges in those latter tw Figure 7 shows a very clear peak in rainfall immediately following the early May, July, and September cloudbus oeprations. The early August cloudbusting operation took place during the normal, rainy summer monsoon of th Colorado River Basin region, and so its influences were lost in the background chaos of the normal August rainf early June operation was followed by only a very small rainfall episode.

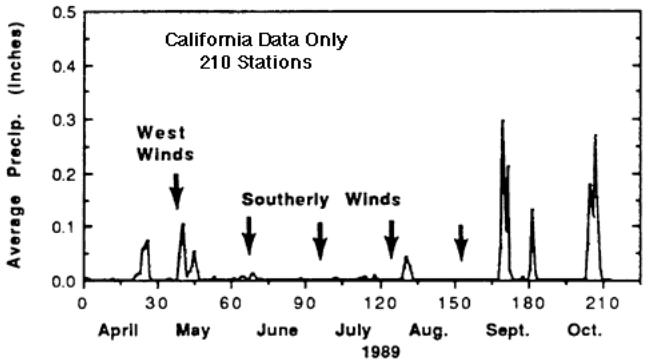

Figure 6: Averaged Daily Precipitation from 210 rainfall measuring stations in Southern California. The d five different cloudbusting operations are marked with an arrow. Only the first of these operations was cha by strong westerly winds and moisture from the Pacific Ocean, and only this first operation yielded signific California.

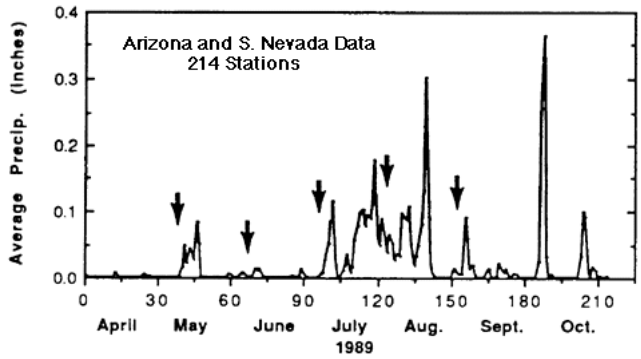

Figure 7: Averaged Daily Precipitation from 214 rainfall measuring stations in Arizona and Southern Nev southerly winds and moisture brought rains to this area in the last four of the cloudbusting operations (marl arrow), even though rains did not develop at those times in Southern California.

Figure 8 shows the daily rainfall data from all 424 rain gauges in the three state regions combined. A periodicity atmospheric pulsation similar to that seen in wetland regions, was observed in the data for the months of Septem October, and persisted for several weeks after the last of the cloudbusting operations. However, no attempt has b find out how often this pulsatory rainfall characteristic naturally develops in this large dryland basin region. The indicate that four out of the five cloudbusting operations were followed by significant peaks in rainfall, which is the success rate (80%) observed in prior cloudbusting operations (11).

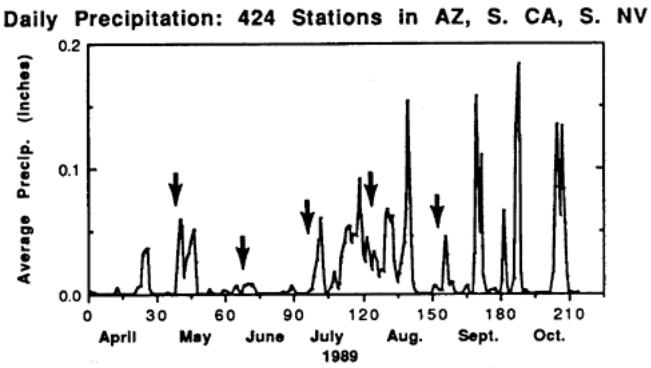

Figure 8: Averaged Daily Precipitation from 424 rainfall measuring stations in Arizona, Southern Californ Southern Nevada combined. Peaks in rainfall follow closely after the cloudbusting operations, marked witl A regular pulsation of rainfall is apparent in September and October.

The averaged daily precipitation values for all 424 rain gauges in the entire three-state region were converted to of-30-day maximum values. This was done for each of the five individual cloudbusting operations, centered on t of each operation, such that daily precipitation percentage values for two weeks before and after each operation

obtained. The percentage data preceding and following the five separate cloudbusting operations were then over averaged together. This approach demonstrated an even more solid connection between thefive 1989 cloudbustir operations and rainfall in the arid Colorado River Basin. As seen in Figure 9, the rainfall occurring within the ba after the cloudbusting operations was approximately twice that observed during the pre-operations period. This p doubled rainfall developed about 48 hours after the onset of operations and persisted for approximately one weel

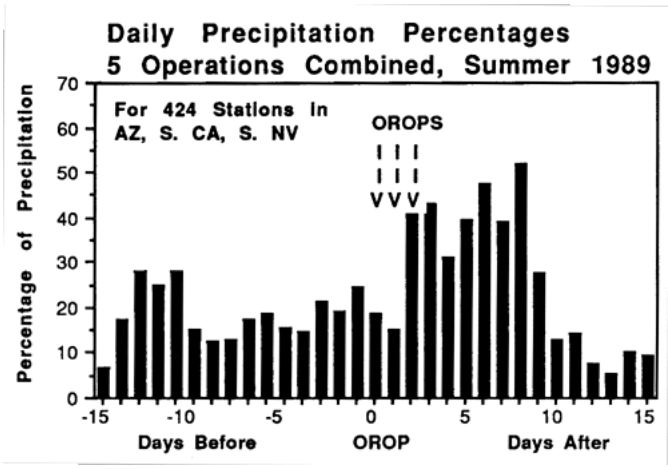

Figure 9: Averaged Daily Precipitation Percentages for the five cloudbusting operations combined, as ove centered on the first day of each operation. A clear and significant increase in precipitation begins about 48 the onset of cloudbusting operations, lasting for approximately one week thereafter. This increase in rains i double the amount observed during the two-week period prior to operations, or during the last week in the period.

The implications of these atmospheric responses to the influence of a single large cloudbuster working for only a of each month are startling, and their importance during an epoch of expanding deserts is significant. While thes results are very encouraging, much remains to be learned about the atmospheric-energetic processes involved in and maintenance of this particular desert region. The lessons learned from this region may or may not be fully ap other desert regions. Also, many questions were raised about the potential ecological and social effects of greeni desert region through use of the cloudbuster. Desert greening is *climate change*, after all, and this implies change fauna, and human habitat. Decisions on any major undertaking to bring about climate change, even apparently b change, must be made on a social level. However, unplanned climate change is already occurring in the reckless assault on the Earth's forests and grasslands by nearly every human culture around the world and in the related a human-forced expansion of the Earth's deserts. The drylands and deserts of the world do not exist in a stable equ with respect to surrounding wetland regions; the deserts are getting larger by the year, while the green areas of tl shrinking in size (12). It is doubtful that any rational collective social decisions can be made regarding these new findings until at least the irrational, antisocial destruction the Earth's living plant cover has been halted.

These important concerns aside, one thing is very clear: The data from this experiment clearly support Dr. Reich from the 1950s that *the cloudbuster possesses a true desert greening capability* (13). Certainly, an extended peri research is warranted.*

* Throughout the 1990s, extended periods of field research were undertaken with the Reich cloudbuster by DeMeo, in the harsh deserts of Israel, Namibia, and the Eastern Sahel of Africa. Details on those additional experiments have been published in various issues of *Pulse of the Planet* journal, available at: http://www.orgonelab.org/cart/xpulse.htm A full Bibliography of Dr. DeMeo's publications is also posted t site. A short listing of his major publications on cloudbusting is given below.

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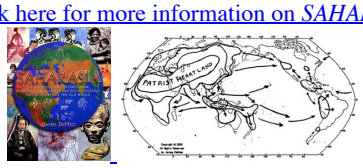

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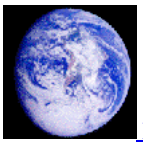
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